To determine an infinite series is convergent or divergence, we may use some of the following tests:

Integral test Direct comparison test Limit comparison test Ratio test Root test	Series that have **NO NEGATIVE** terms.	The *n*-the term test for divergence Alternating series test Absolute convergence test	Series that may have **SOME NEGATIVE** terms.

9.3 The Integral test

Give a series $\sum a_n$, we have two questions.
1. Does the series $\sum a_n$ converge?
2. If it converges, what is its sum?

We begin to study this question, starting with series that have nonnegative terms. The reason for this restriction is that the partial sums of these series form nondecreasing sequences, and nondecreasing sequences that are bounded from above always converge.
If we establish that a given series does converge, we generally do not have a formula available for its sum. In Section 9.3 – 9.6, we study several tests that enable us to determine whether a series is convergent or divergent without finding its sum.

Nondecreasing Partial Sums

Suppose that $\sum a_n$ is an infinite series with $a_n \geq 0$ for all n. Then each partial sum is greater than or equal to its predecessor because $s_{n+1} = s_n + a_n$, so

$$s_1 \leq s_2 \leq s_3 \leq \cdots \leq s_n \leq s_{n+1} \leq \cdots.$$

Since the partial sums form a nondecreasing sequence, the Monotonic Sequence Theorem (Section 9.1) gives the following result.

Corollary 1:
A series $\sum a_n$ of nonnegative terms converges if and only if its partial sums are bounded from above.

Example 1:
Consider the **harmonic series** $\sum a_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$
Although the *n*th term $1/n \to \infty$ as $n \to \infty$, but the series diverges because there is no upper bound for its partial sums. To see why, group the terms of the series in the following way:

$$1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{> \frac{2}{4} = \frac{1}{2}} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{> \frac{4}{8} = \frac{1}{2}} + \underbrace{\left(\frac{1}{9} + \frac{1}{10} + \cdots + \frac{1}{16}\right)}_{> \frac{8}{16} = \frac{1}{2}} + \cdots.$$

The sum of the first two terms is 1.5. The sum of the next two terms is $\frac{1}{3} + \frac{1}{4}$, which is greater than $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. The sum of the next four terms is $\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}$, which is greater than $\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$. The sum of the next eight terms is $\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16}$, which is greater than 8/16 = 1/2. The sum of the next 16 terms is greater than 16/32 = 1/2, and so on. In general, the sum of 2^m terms ending with $\frac{1}{2^m} + 1$ is greater than $\frac{2^m}{2^{m+1}} = \frac{1}{2}$. Therefore, if $n = 2^k$, then the partial sum s_n is greater than $k/2$, so he sequence of partial sums is not bounded from above. The harmonic series diverges.

Theorem 1 (The Integral Test):
Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_x = f(x)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$ (N a positive integer). Then the series $\sum a_n$ and the integral $\int f(x)dx$ both converge or both diverge.

Take note that to apply the integral test,
1. a_n must be positive and decreasing $\forall\ n \geq N$;
2. Define $f(x) = a_x$, a function which is positive, continuous and decreasing $\forall\ x \geq N$;
3. Evaluate $\int_N^\infty f(x)dx$.
4. $\int_N^\infty f(x)dx$ converges (diverges) $\Rightarrow$ $\sum_{n=N}^\infty a_n$ converges (diverges)

Example 2: Show that the harmonic series $\sum_{n=1}^\infty \frac{1}{n}$ is divergent.

Solution:
Let $f(x) = \frac{1}{x}$, $x \geq 1$. The function is positive, continuous and decreasing for all $x \geq 1$.

$$\int_1^\infty \frac{1}{x}dx = \lim_{b\to\infty} \int_1^b \frac{1}{x}dx = \lim_{b\to\infty} \ln|x|\Big]_1^b = \lim_{b\to\infty} (\ln b - \ln 1) = \infty.$$

$\therefore$ The improper integral $\int_1^\infty \frac{1}{x}dx$ diverges. By the Integral Test, the harmonic series $\sum_{n=1}^\infty \frac{1}{n}$ diverges.

Example 3: The *p*-series
Show that for a real constant p, **the *p*-series** $\sum_{n=1}^\infty \frac{1}{n^p}$ converges if $p > 1$, diverges if $p \leq 1$.

Solution:
If $p > 1$, then $f(x) = \frac{1}{x^p}$ is a positive, continuous, and decreasing function of x. Since

$$\begin{aligned} \int_1^\infty \frac{1}{x^p}dx &= \int_1^\infty x^{-p}dx = \lim_{b\to\infty}\left[\frac{x^{-p+1}}{-p+1}\right]_1^b && \text{Evaluate the improper in} \\ &= \frac{1}{1-p}\lim_{b\to\infty}\left(\frac{1}{b^{p-1}} - 1\right) \\ &= \frac{1}{1-p}(0-1) = \frac{1}{p-1}, && b^{p-1} \to \infty \text{ as } b \to \infty \\ & && \text{because } p-1 > 0. \end{aligned}$$

the series converges by the Integral Test.
If $p \leq 0$, the series diverges by the nth-term test.
If $0 < p < 1$, then $1 - p > 0$ and
$\int_{x=1}^\infty \frac{1}{x^p}dx = \lim_{b\to\infty} \int_{x=1}^b \frac{1}{x^p}dx = \lim_{b\to\infty} \frac{x^{1-p}}{1-p}\Big]_1^b = \frac{1}{1-p}\lim_{b\to\infty}[b^{1-p} - 1] = \infty$. Therefore, the series diverges by the Integral Test.
If $p = 1$, we have the (divergent) harmonic series $\sum_{n=1}^\infty \frac{1}{n}$.
In summary, we have convergence for $p > 1$ but divergence for all other values of p.

Remark:

Take note that, when $\int_N^\infty f(x)dx$ and $\sum_{n=N}^\infty a_n$ both converge, **DOES NOT mean they have the same value.**

Example 4: Does the series $\sum_{n=1}^\infty \frac{1}{(n+1)^2}$ converge?

Solution:

Let $f(x)=\frac{1}{(x+1)^2}$. The function is positive, continuous and decreasing for $x \geq 1$.

$\int_1^\infty \frac{1}{(x+1)^2}dx = \lim_{b\to\infty}\int_1^b \frac{1}{(x+1)^2}dx = \lim_{b\to\infty}\left.\frac{-1}{x+1}\right]_1^b = \lim_{b\to\infty}\left(-\frac{1}{b+1}+\frac{1}{2}\right)=\frac{1}{2}$. $\therefore$ The improper integral converges.

By the Integral Test, the series $\sum_{n=1}^\infty \frac{1}{(n+1)^2}$ converges.

Take note that $\sum_{n=1}^\infty \frac{1}{(n+1)^2}=\frac{\pi^2}{6}-1 \neq \int_1^\infty \frac{1}{(x+1)^2}dx=\frac{1}{2}$.

Example 5: Determine the convergence or divergence of the series.

(a) $\sum_{n=1}^\infty ne^{-n^2}$ (b) $\sum_{n=1}^\infty \frac{1}{2^{\ln n}}$

Solution:

(a) We apply the Integral Test and find that

$$\int_1^\infty \frac{x}{e^{x^2}}dx = \frac{1}{2}\int_1^\infty \frac{du}{e^u} \qquad u = x^2,\ du = 2x\,dx$$

$$= \lim_{b\to\infty}\left[-\frac{1}{2}e^{-u}\right]_1^b$$

$$= \lim_{b\to\infty}\left(-\frac{1}{2e^b}+\frac{1}{2e}\right)=\frac{1}{2e}.$$

Since the integral converges, the series also converges.

(b) Again applying the Integral Test,

$$\int_1^\infty \frac{dx}{2^{\ln x}} = \int_0^\infty \frac{e^u\,du}{2^u} \qquad u = \ln x,\ x = e^u,\ dx = e^u\,du$$

$$= \int_0^\infty \left(\frac{e}{2}\right)^u du$$

$$= \lim_{b\to\infty}\frac{1}{\ln\left(\frac{e}{2}\right)}\left(\left(\frac{e}{2}\right)^b - 1\right) = \infty. \qquad (e/2) > 1$$

The improper integral diverges, so the series diverges also.

Error Estimation

Bounds for the Remainder in the Integral Test

Suppose $\{a_k\}$ is a sequence of positive terms with $a_k = f(k)$, where f is a continuous positive decreasing function of x for all $x \geq n$, and that Σa_n converges to S. Then the remainder $R_n = S - s_n$ satisfies the inequalities

$$\int_{n+1}^{\infty} f(x)\,dx \leq R_n \leq \int_{n}^{\infty} f(x)\,dx. \qquad (1)$$

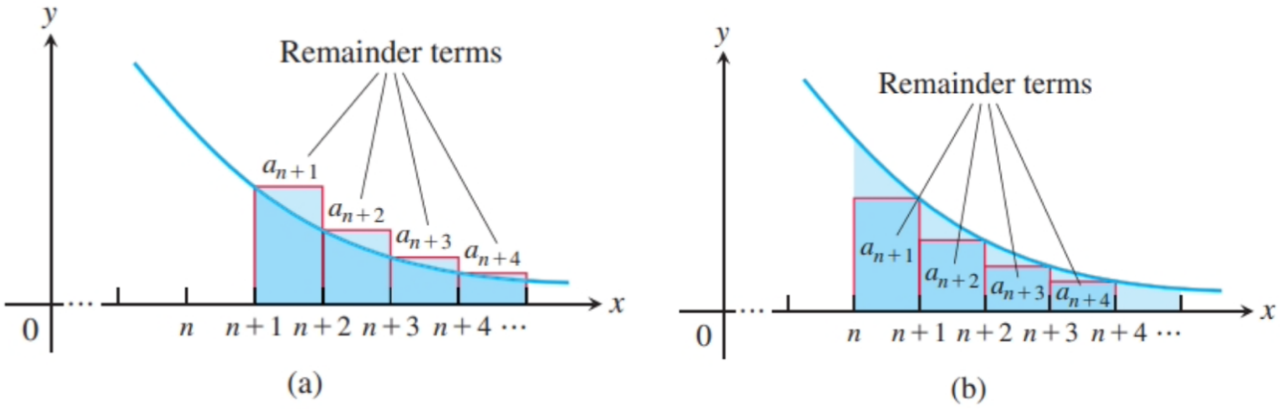


Figure 1 The remainder when using n terms in (a) larger than the integral of f over $[n + 1, \infty)$. (b) smaller than the integral of f over $[n, \infty)$.

Example 6: Estimate the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ using the inequalities in (2) and $n = 10$.

Solution:

$$\int_{n}^{\infty} \frac{1}{x^2}dx = \lim_{b\to\infty}\left[-\frac{1}{x}\right]_n^b = \lim_{b\to\infty}\left(-\frac{1}{b} + \frac{1}{n}\right) = \frac{1}{n}.$$

Using this result with the inequalities in (2), we get

$$s_{10} + \frac{1}{11} \leq S \leq s_{10} + \frac{1}{10}.$$

Taking $s_{10} = 1 + (1/4) + (1/9) + (1/16) + \cdots + (1/100) \approx 1.54977$, these last inequalities give $1.64068 \leq S \leq 1.64977$.

If we approximate the sum S by the midpoint of this interval, we have $\sum_{n=1}^{\infty} \frac{1}{n^2} \approx 1.6452$.

The error in this approximation is then less than half the length of the interval, so the error is less than 0.005.

Using a trigonometric *Fourier series* (Section 19.4) that S is equal to $\pi^2/6 \approx 1.64493$.

9. 4 Comparison Test

- To compare $\sum a_n$ with $\sum b_n$ that is **known to be convergent or divergent**.
- May apply to series which has a form ***similar*** to geometric or *p*-series. When a_n is a rational or algebraic (involving roots of polynomial) function, should be compared with *p*- series.

(1) Direct Comparison Test

Direct Comparison Test:
Let $\sum a_n$ be a series with no negative terms.

(a) $0 \le a_n \le b_n \ \forall\ n > N$ and $\sum b_n$ converges $\Rightarrow \sum a_n$ converges.

(b) $0 \le b_n \le a_n \ \forall\ n > N$ and $\sum b_n$ diverges $\Rightarrow \sum a_n$ diverges.

Note: **DO NOT** apply the direct comparison test to test Σa_n when

(1) $a_n \le b_n \ \forall\ n > N$ and Σb_n diverges, or (2) $a_n \ge b_n \ \forall\ n > N$ and Σb_n converges.

Example 1:
Test the following series by the direct comparison test.

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{5}{5n-1}$ (2) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} + \cdots$

(3) $\displaystyle 5 + \frac{2}{3} + \frac{1}{7} + 1 + \frac{1}{2+\sqrt{1}} + \frac{1}{4+\sqrt{2}} + \frac{1}{8+\sqrt{3}} + \cdots + \frac{1}{2^n+\sqrt{n}} + \cdots$

Solution:

(1) Since $0 < \dfrac{1}{n} < \dfrac{5}{5n-1} \ \forall\ n \ge 1$, and $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$ is the Harmonic series which is a divergent series, by the direct comparison test, $\displaystyle\sum_{n=1}^{\infty} \frac{5}{5n-1}$ diverges.

(2) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \sum_{n=2}^{\infty} \frac{1}{n!}.$

$n! = 1(2)(3)\cdots(n) \ge 1(2)(2)\cdots(2) = 2^{n-1} \ \forall\ n \ge 2.$ Hence, $0 < \dfrac{1}{n!} \le \dfrac{1}{2^{n-1}} \ \forall\ n \ge 2.$

$\displaystyle\sum_{n=2}^{\infty} \frac{1}{2^{n-1}}$ is a convergent geometric series with $a = \frac{1}{2}$ and $|r| = \frac{1}{2} < 1.$

By the direct comparison test, $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n!}$ converges, and hence $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n!} = 2 + \sum_{n=2}^{\infty} \frac{1}{n!}$ also converges.

Take note that $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \sum_{n=2}^{\infty} \frac{1}{n!} \le 2 + \sum_{n=2}^{\infty} \frac{1}{2^{n-1}} = 2 + \frac{\frac{1}{2}}{1-\frac{1}{2}} = 3.$ This fact that 3 is an upper bound for the partial sums of $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n!}$ does not mean that the series converges to 3.

(3) $5+\frac{2}{3}+\frac{1}{7}+1+\frac{1}{2+\sqrt{1}}+\frac{1}{4+\sqrt{2}}+\frac{1}{8+\sqrt{3}}+\cdots+\frac{1}{2^n+\sqrt{n}}+\cdots=\frac{143}{21}+\sum_{k=1}^{\infty}\frac{1}{2^k+\sqrt{k}}$.

$0<\frac{1}{2^k+\sqrt{k}}<\frac{1}{2^k}\ \forall\ k\geq 1$, and $\sum_{k=1}^{\infty}\frac{1}{2^k}$ is a convergent geometric series with $a=\frac{1}{2}$ and $|r|=\frac{1}{2}<1$, by the direct comparison test, $\sum_{k=1}^{\infty}\frac{1}{2^k+\sqrt{k}}$ converges.

Hence, $5+\frac{2}{3}+\frac{1}{7}+1+\frac{1}{2+\sqrt{1}}+\frac{1}{4+\sqrt{2}}+\frac{1}{8+\sqrt{3}}+\cdots+\frac{1}{2^n+\sqrt{n}}+\cdots=\frac{143}{21}+\sum_{k=1}^{\infty}\frac{1}{2^k+\sqrt{k}}$ converges.

Example 2:

Test $\sum_{n=1}^{\infty}\frac{1}{n+3^n}$ by the direct comparison test.

Solution:

A suitable comparison is $0<\frac{1}{n+3^n}<\frac{1}{3^n}\ \forall\ n\geq 1$, and $\sum_{n=1}^{\infty}\frac{1}{3^n}$ is a convergent geometric series.

By the direct comparison test, $\sum_{n=1}^{\infty}\frac{1}{n+3^n}$ converges.

Although $0<\frac{1}{n+3^n}<\frac{1}{n}\ \forall\ n\geq 1$, DO NOT apply the direct comparison test with $\sum_{n=1}^{\infty}b_n=\sum_{n=1}^{\infty}\frac{1}{n}$, because $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges.

(2) Limit Comparison Test:

The Limit Comparison Test:

Suppose that $a_n > 0$ and $b_n > 0$ and for all $n \geq N$ (N an integer)

(1) $\lim_{n\to\infty}\frac{a_n}{b_n}=c>0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.

(2) $\lim_{n\to\infty}\frac{a_n}{b_n}=0$ and $\sum b_n$ converges, then $\sum a_n$ converges.

(3) $\lim_{n\to\infty}\frac{a_n}{b_n}=\infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Note: **DO NOT** apply the Limit Comparison Test to test $\sum a_n$ when

(1) $\lim_{n\to\infty}\frac{a_n}{b_n}=0$ and $\sum b_n$ diverges, or (2) $\lim_{n\to\infty}\frac{a_n}{b_n}=\infty$ and Σb_n converges.

Example 3:
Which of the following series converge, and which diverge?

(1) $\frac{3}{4}+\frac{5}{9}+\frac{7}{16}+\frac{9}{25}+\cdots=\sum_{n=1}^{\infty}\frac{2n+1}{(n+1)^2}=\sum_{n=1}^{\infty}\frac{2n+1}{n^2+2n+1}$

(2) $\frac{1}{1}+\frac{1}{3}+\frac{1}{7}+\frac{1}{15}+\cdots=\sum_{n=1}^{\infty}\frac{1}{2^n-1}$

(3) $\frac{1+2\ln 2}{9}+\frac{1+3\ln 3}{14}+\frac{1+4\ln 4}{21}+\cdots=\sum_{n=2}^{\infty}\frac{1+n\ln n}{n^2+5}$

Solution:

(1) Let $a_n = (2n+1)/(n^2+2n+1)$. For large n, we expect a_n to behave like $2n/n^2 = 2/n$ since the leading terms dominate for large n, so we let $b_n = 1/n$. The numbers a_n and b_n are positive for each n, the series $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges and

$$\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{2n+1}{n^2+2n+1}(n)=\lim_{n\to\infty}\frac{(2n^2+n)\left(\frac{1}{n^2}\right)}{(n^2+2n+1)\left(\frac{1}{n^2}\right)}$$

$$=\lim_{n\to\infty}\frac{2+\frac{1}{n}}{1+\frac{2}{n}+\frac{1}{n^2}}=2>0$$

By the limit comparison test, $\sum_{n=1}^{\infty}\frac{2n^2+n}{n^2+2n+1}$ diverges.

(2) Let $a_n = 1/(2^n-1)$. For large n, we expect a_n to behave like $1/2^n$, so we let $b_n = 1/2^n$. The numbers a_n and b_n are positive for each n, the series $\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n$ is a convergent geometric series with $|r| = ½ < 1$.

$$\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{1}{2^n-1}(2^n)=\lim_{n\to\infty}\frac{(2^n)\left(\frac{1}{2^n}\right)}{(2^n-1)\left(\frac{1}{2^n}\right)}$$

$$=\lim_{n\to\infty}\frac{1}{1-\frac{1}{2^n}}=\frac{1}{1-0}=1>0$$

By the limit comparison test, $\sum_{n=1}^{\infty}\frac{1}{2^n-1}$ converges.

(3) Let $a_n = (1+n\ln n)/(n^2+5)$. For large n, we expect a_n to behave like $n\ln n/n^2$ which is greater than $1/n$ for n >= 3, so we let $b_n = 1/n$. The numbers a_n and b_n are positive for each n, the series $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges and

$$\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{1+n\ln n}{n^2+5}(n)=\infty.$$

By the limit comparison test, $\sum_{n=1}^{\infty}\frac{1+n\ln n}{n^2+5}$ diverges.

Example 4:
Determine which the following series is convergent and which is divergent.

(1) $\sum_{n=1}^{\infty} \frac{2}{\sqrt{n}+\sqrt[3]{n}}$ (2) $\sum_{n=1}^{\infty} \frac{n+1}{n\sqrt{n^4-1}}$ (3) $\sum_{n=1}^{\infty} \frac{\sin^2 n}{n^2}$

Solution:

(1) $\forall\ n \geq 1,\ 0 \leq \sqrt[3]{n} \leq \sqrt{n} \ \Rightarrow\ 0 \leq \sqrt[3]{n}+\sqrt{n} \leq \sqrt{n}+\sqrt{n} \ \Rightarrow\ 0 \leq \frac{1}{\sqrt{n}} \leq \frac{2}{\sqrt[3]{n}+\sqrt{n}}$.

Furthermore, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is a divergent *p*-series with $p = ½ < 1$.

By the direct comparison test, $\sum_{n=1}^{\infty} \frac{2}{\sqrt{n}+\sqrt[3]{n}}$ diverges.

Alternative method (The Limit Comparison Test):

Let $a_n = \frac{2}{\sqrt{n}+\sqrt[3]{n}}$. Take $b_n = \frac{1}{\sqrt{n}}$.

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is a divergent *p*-series, and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{2\sqrt{n}}{\sqrt{n}+\sqrt[3]{n}} = \lim_{n\to\infty} \frac{2\sqrt{n}}{\sqrt{n}\left(1+\frac{1}{\sqrt[6]{n}}\right)} = \lim_{n\to\infty} \frac{2}{\left(1+\frac{1}{\sqrt[6]{n}}\right)} = 2 > 0.$$

By the limit comparison test, $\sum_{n=1}^{\infty} \frac{2}{\sqrt{n}+\sqrt[3]{n}}$ diverges.

(2) Let $a_n = \frac{n+1}{n\sqrt{n^4-1}}$ Take $b_n = \frac{1}{n^2}$.

$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ is a convergent *p*-series with $p = 2 > 1$,

And $\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{(n+1)n^2}{n\sqrt{n^4-1}} = \lim_{n\to\infty} \frac{1+(1/n)}{\sqrt{1-(1/n^4)}} = 1 > 0 \cdot$

By the limit comparison test, $\sum_{n=1}^{\infty} \frac{n+1}{n\sqrt{n^4-1}}$ converges.

(3) $0 \leq \frac{\sin^2 n}{2^n} \leq \frac{1}{2^n}$ $\forall\ n \geq 1$, and $\sum_{n=1}^{\infty} \frac{1}{2^n}$ is a convergent geometric with $|r| = ½ < 1$.

By the direct comparison test, $\sum_{n=1}^{\infty} \frac{\sin^2 n}{2^n}$ converges.

Example 5: Does $\sum_{n=1}^{\infty} \frac{\ln n}{n^{3/2}}$ converge?

Solution:

When n sufficiently large, $\ln n$ grows more slowly that $n^c \ \forall\ c > 0$. $\because \ \lim_{x\to\infty} \frac{\ln x}{x^c} = 0 \ \ \forall\ c > 0.$

Take $a_n = \frac{\ln n}{n^{3/2}}$ and $b_n = \frac{1}{n^{5/4}}$, we have

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\ln n}{n^{\frac{3}{2}}}\left(n^{\frac{5}{4}}\right) = \lim_{n\to\infty} \frac{\ln n}{n^{\frac{1}{4}}} = \lim_{x\to\infty} \frac{\ln x}{x^{\frac{1}{4}}} \quad \text{where } f(x) = \frac{\ln x}{x^{\frac{1}{4}}} \text{ with real } x \geq 1$$

$$= \lim_{x\to\infty} \frac{1/x}{x^{-\frac{3}{4}}/4} = \lim_{x\to\infty} \frac{4}{x^{\frac{1}{4}}} = 0.$$

Since $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{5/4}}$ is a p-series with p > 1, it converges, so $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{\ln n}{n^{3/2}}$ converges by Limit Comparison Test.

Remark: For $\sum_{n=1}^{\infty} \frac{\ln n}{n^c}$, to apply the Limit Comparison Test, we can choose any p where $1 < p < c$ so that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges and $\lim_{n\to\infty} \frac{a_n}{b_n} = 0.$

Let $a_n = \frac{\ln n}{n^c}$, $c < 0$ and $b_n = \frac{1}{n^p}$, we have

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\ln n}{n^c}\left(n^p\right) = \lim_{n\to\infty} \frac{\ln n}{n^{c-p}} = \lim_{x\to\infty} \frac{\ln x}{x^{c-p}} \quad \text{where} \quad f(x) = \frac{\ln x}{x^c} \text{ with real } x \geq 1$$

$$= \lim_{x\to\infty} \frac{1/x}{(c-p)x^{c-p-1}} = \lim_{x\to\infty} \frac{1}{(c-p)x^{c-p}}.$$

Clearly, $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ if $c - p > 0$.

In this case, to apply the Limit Comparison Test, we have to choose a **convergent** $\sum b_n$, that is a convergent p-series with $p > 1$. So, we can choose any p where $1 < p < c$, , for example $p = \frac{5}{4}$ in Example 4.

(Take note that, $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ when $c - p < 0$ and $\sum b_n$ is a convergent p-series with $p > c$. We can't apply the limit comparison test with this choice of $\sum b_n$.)